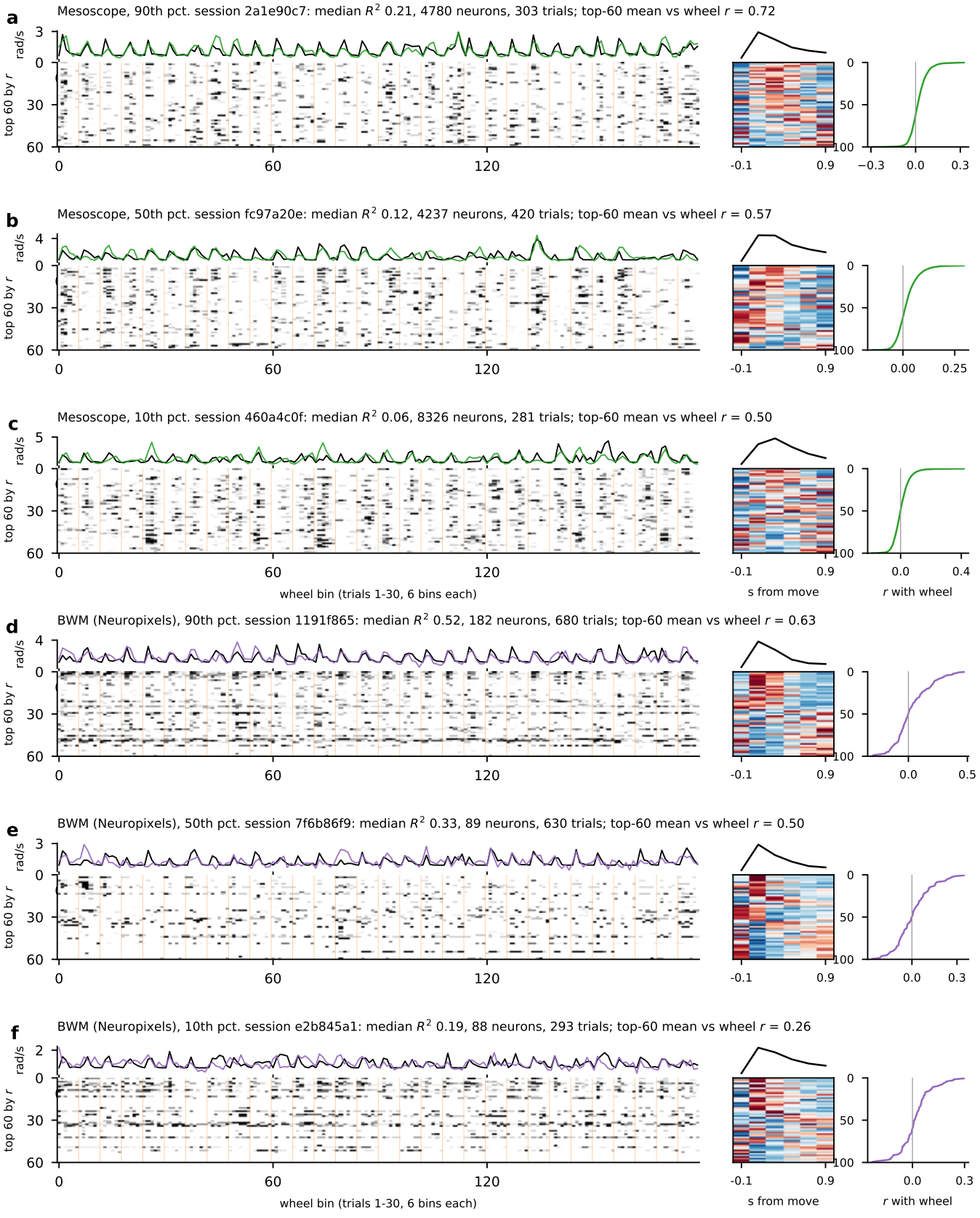


Figure 1: Neural correlates of wheel speed. Each row: wheel speed (top; rad/s) in the 6 wheel bins (0.2 s, -0.2 to 1 s around first movement) of the first 30 trials, concatenated (orange lines: trial boundaries; bins are not contiguous across trials; coloured: mean z-scored activity of the neurons below, rescaled), above the z-scored activity of the 60 neurons most positively correlated with wheel speed (sorted by r, highest on top). Neurons are ranked by r over all bins of the remaining trials, so the shown trials are held out from the selection. Middle: trial-averaged activity of all neurons of the session per bin (z-scored per neuron, same order), with the mean wheel speed above. Right: held-out-trial r of all neurons against their rank (%). Sessions at the 90th, 50th and 10th percentile of the median R^2 over bins (all neurons per 0.4 mm bin, revised decoder with lagged features; sessions with $\geq$30 neurons). **a** Mesoscope session 2a1e90c7 (90th percentile of the session median wheel-speed R^2): 95th percentile of r over neurons 0.12, maximum 0.32, 7% of neurons with $r > 0.1$; mean of the top 60 neurons vs wheel speed in the shown trials r = 0.72. **b** Mesoscope session fc97a20e (50th percentile of the session median wheel-speed R^2): 95th percentile of r over neurons 0.11, maximum 0.29, 6% of neurons with $r > 0.1$; mean of the top 60 neurons vs wheel speed in the shown trials r = 0.57. **c** Mesoscope session 460a4c0f (10th percentile of the session median wheel-speed R^2): 95th percentile of r over neurons 0.09, maximum 0.42, 4% of neurons with $r > 0.1$; mean of the top 60 neurons vs wheel speed in the shown trials r = 0.50. **d** BWM (Neuropixels) session 1191f865 (90th percentile of the session median wheel-speed R^2): 95th percentile of r over neurons 0.31, maximum 0.47, 26% of neurons with $r > 0.1$; mean of the top 60 neurons vs wheel speed in the shown trials r = 0.63. **e** BWM (Neuropixels) session 7f6b86f9 (50th percentile of the session median wheel-speed R^2): 95th percentile of r over neurons 0.23, maximum 0.35, 20% of neurons with $r > 0.1$; mean of the top 60 neurons vs wheel speed in the shown trials r = 0.50. **f** BWM (Neuropixels) session e2b845a1 (10th percentile of the session median wheel-speed R^2): 95th percentile of r over neurons 0.19, maximum 0.30, 15% of neurons with $r > 0.1$; mean of the top 60 neurons vs wheel speed in the shown trials r = 0.26.